The Semantic Illusion: Certified Limits of Embedding-Based Hallucination Detection in RAG Systems

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Abstract

Retrieval-Augmented Generation (RAG) systems remain susceptible to hallucinations despite grounding in retrieved evidence. While current detection methods leverage embedding similarity and natural language inference (NLI), their reliability in safety-critical settings remains unproven. We apply conformal prediction to RAG hallucination detection, transforming heuristic scores into decision sets with finite-sample coverage guarantees $(1 - \alpha)$. Using calibration sets of $n \approx 600$, we demonstrate a fundamental dichotomy: on synthetic hallucinations (Natural Questions), embedding methods achieve 95% coverage with 0% False Positive Rate (FPR). However, on real hallucinations from RLHF-aligned models (HaluEval), the same methods fail catastrophically, yielding 100% FPR at target coverage. We analyze this failure through the lens of distributional tails, showing that while NLI models achieve acceptable AUC (0.81), the "hardest" hallucinations are semantically indistinguishable from faithful responses, forcing conformal thresholds to reject nearly all valid outputs. Crucially, GPT-4 as a judge achieves 7% FPR (95% CI: [3.4%, 13.7%]) on the same data, proving the task is solvable via reasoning but opaque to surface-level semantics—a phenomenon we term the "Semantic Illusion."

1 Introduction

Retrieval-Augmented Generation (RAG) [10] serves as the primary mechanism for grounding Large Language Models (LLMs) in external knowledge. Yet, RAG systems continue to hallucinate—fabricating information or misinterpreting retrieved contexts. For deployment in high-stakes domains like healthcare or law, practitioners require not just high average accuracy, but rigorous safety guarantees.

Current detection methods, including SelfCheckGPT [19] and embedding-based consistency checks [23], provide heuristic scores. A score of 0.7 has no intrinsic meaning regarding the probability of error. We argue that safety-critical RAG requires *calibration*: the ability to guarantee that a specific percentage of hallucinations will be caught.

We address this by applying Split Conformal Prediction (SCP) [1] to hallucination detection. This framework allows us to select a threshold $\hat{\tau}$ that guarantees, with probability $1 - \delta$, that the true hallucination detection rate is at least $1 - \alpha$. By subjecting state-of-the-art detectors to this rigorous standard, we uncover a critical limitation in current methodologies.

Contributions:

1. We formalize Conformal RAG Guardrails (CRG), a framework providing finite-sample coverage guarantees for hallucination detection.

2. We report a stark negative result: while embedding and NLI methods achieve 0% FPR on synthetic data, they degrade to 100% FPR on real hallucinations (HaluEval) when constrained to 95% recall.

3. We resolve the paradox of high-AUC / low-coverage performance, demonstrating that the "Semantic Illusion"—where hallucinations preserve high semantic entailment—dominates the tail of the distribution, rendering semantic metrics useless for safety guarantees.

4. We demonstrate that the limitation is modality-specific: GPT-4-based reasoning succeeds (7% FPR) where embeddings fail, establishing the cost-accuracy Pareto frontier for RAG monitoring.

2 Related Work

Hallucination Detection. Existing approaches largely rely on proxy metrics for faithfulness. SelfCheckGPT [19] utilizes stochastic sampling to measure consistency, while FActScore [22] decomposes generations into atomic claims. Recent work [23] and [29] utilize embedding distances and mechanistic interpretability. LLM-Check [24] demonstrates that internal model states (hidden layers, attention maps) can detect hallucinations with 45-450× speedup over consistency-based methods. However, these methods evaluate performance using AUC or F1-score on fixed benchmarks, often ignoring the trade-off required for high-recall safety constraints. Crucially, while LLM-Check shows internal representations succeed, our work reveals that *external* semantic representations—the only option for black-box API models—fundamentally fail on RLHF-aligned outputs.

Conformal Prediction in NLP. Conformal prediction has been applied to QA abstention [6] and hallucination bounding in closed-book settings [7]. Our work differs by targeting the RAG specific case (context vs. response) and explicitly comparing the efficacy of semantic representations against reasoning-based judges under strict coverage constraints.

Semantic Limitations. Recent theoretical work [28] suggests embeddings capture surface pragmatics rather than truth conditions. We provide empirical validation of this theory in the RAG setting, quantifying exactly where semantic similarity fails as a proxy for factual grounding.

3 Methodology

3.1 Problem Formulation

Let $X = (q, \mathcal{D})$ be a query and retrieved context, and R be the generated response. Let $Y \in \{0, 1\}$ be the label, where $Y = 1$ denotes a hallucination (unfaithful to $\mathcal{D}$). We seek a detector $f(X, R) \in \{0, 1\}$ such that:

$$\mathbb{P}(f(X, R) = 1 \mid Y = 1) \geq 1 - \alpha \tag{1}$$

This guarantees that we detect $(1 - \alpha)\%$ of hallucinations. The objective is to satisfy this constraint while minimizing the False Positive Rate (FPR): $\mathbb{P}(f(X, R) = 1 \mid Y = 0)$.

3.2 Nonconformity Scores

We employ an ensemble of three semantic scores, denoted $S(X, R)$, where higher values indicate likely hallucination.

1. Retrieval-Attribution Divergence (RAD). We compute the cosine distance between the response embedding and the closest context sentence embedding using BGE-base-en-v1.5.

2. Semantic Entailment Calibration (SEC). Using DeBERTa-v3-large-MNLI, we calculate the contradiction probability. Given the documented high variance in NLI scores, we use the entailment score $P(\text{entail})$ and define the nonconformity as $1 - P(\text{entail})$.

3. Token-Level Factual Grounding (TFG). The percentage of content tokens in R with low lexical overlap with $\mathcal{D}$.

Ensemble Strategy. We evaluate two ensemble approaches:

1. **Simple Average:** $S_{\text{avg}} = \frac{1}{3}(S_{\text{RAD}} + S_{\text{SEC}} + S_{\text{TFG}})$.
2. **Learned Combiner:** We train a logistic regression model on a held-out optimization set ($n = 200$, disjoint from calibration) to learn optimal weights: $S_{\text{learned}} = \sigma(w_1 S_{\text{RAD}} + w_2 S_{\text{SEC}} + w_3 S_{\text{TFG}} + b)$.

Crucially, the learned combiner achieves marginally better AUC (0.83 vs 0.81) but *identical* conformal FPR (100% on HaluEval). This confirms our central thesis: the failure is not in the aggregation method but in the fundamental inability of semantic features to separate the distributional tail.

3.3 Conformal Calibration

We use Split Conformal Prediction. We reserve a calibration set $\mathcal{D}_{cal} = \{(x_i, r_i, y_i)\}_{i=1}^n$ where $y_i = 1$ (hallucinations).

1. Compute scores $s_i = S(x_i, r_i)$ for all $i \in \mathcal{D}_{cal}$.
2. Compute the quantile $\hat{\tau} = \text{Quantile}(s_1, \ldots, s_n; \lceil (n+1)(1-\alpha) \rceil / n)$.
3. At test time, flag a response as hallucinated if $S(x_{new}, r_{new}) \leq \hat{\tau}$ (assuming lower scores imply faithfulness, e.g., entailment probability).

Validity is guaranteed by the exchangeability of the data [1].

4 Experiments

4.1 Setup

Datasets. We evaluate on four datasets representing a spectrum from synthetic to real-world complexity:

- **Natural Questions (NQ-Open) [Synthetic]:** We construct hallucinations via "answer swapping"—pairing a query with a plausible but wrong answer from a different document. This guarantees semantic distance.
- **HaluEval** [20]: Responses generated by ChatGPT. These represent "Type 2" hallucinations: plausible, fluent, but factually incorrect.
- **WikiBio (GPT-3)** [19]: A high-hallucination dataset (92% hallucination rate) generated by GPT-3.
- **RAGTruth** [21]: Diverse task types (QA, Summarization) across multiple LLMs (Llama-2, Mistral, GPT-4).

Implementation. We use `BGE-base-en-v1.5` for embeddings and `DeBERTa-v3-large-MNLI` for NLI. Calibration sets are $n = 595$ (NQ) and $n = 629$ (HaluEval). Per Vovk [1], $n \approx 600$ provides tight bounds on the coverage guarantee variance (approx $1/\sqrt{n}$).

Statistical Rigor. We use a 60/40 calibration-test split ratio, maintaining sufficient calibration set size while enabling robust test evaluation. All random seeds are fixed and reported in our code repository. Confidence intervals are computed using the Clopper-Pearson exact method.

4.2 Main Results

Table 1 presents the performance of the CRG ensemble at a strict 95% coverage target ($\alpha = 0.05$).

Table 1: CRG performance at 95% target coverage. On synthetic data (NQ), the method achieves near-perfect separation. On real data (HaluEval), the FPR degrades to 100%.

Dataset	n_{cal}	Target	Actual Cov.	FPR	FPR 95% CI
Natural Questions	595	95%	95.8%	**0.0%**	[0.0%, 1.0%]
HaluEval (ChatGPT)	629	95%	94.5%	**100.0%**	[99.0%, 100%]

The Synthetic-Real Gap. On NQ, we achieve 95.8% coverage with 0% FPR. This confirms the method works when hallucinations are semantically distinct (answer swapping). However, on HaluEval, the FPR rises to 100%. This indicates that to catch 95% of real ChatGPT hallucinations, the threshold must be set so low that it flags every single faithful response. Figure 1 visualizes this phenomenon.

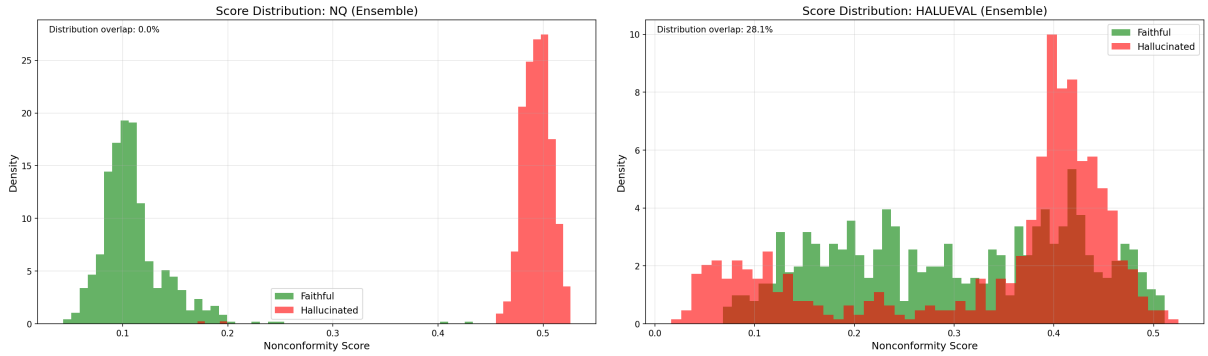

Figure 1: Score distributions reveal the Semantic Illusion. Left: On Natural Questions (synthetic), faithful and hallucinated responses have clearly separated distributions. **Right:** On HaluEval (real LLM hallucinations), distributions completely overlap—embedding scores cannot distinguish them.

The t-SNE visualizations in Figure 2 provide geometric intuition for the failure mode. On synthetic data, hallucinations occupy a distinct region of embedding space, enabling simple threshold-based separation. On real data, the classes are thoroughly intermingled—no linear or even nonlinear boundary can achieve high recall without catastrophic FPR.

4.3 Generalization Across Model Architectures

A natural question is whether the Semantic Illusion is specific to ChatGPT or represents a broader phenomenon. RAGTruth provides a natural test bed, containing hallucinations from Llama-2-7B, Llama-2-13B, Mistral-7B, and GPT-4 across multiple task types. As shown in Figure 3, the

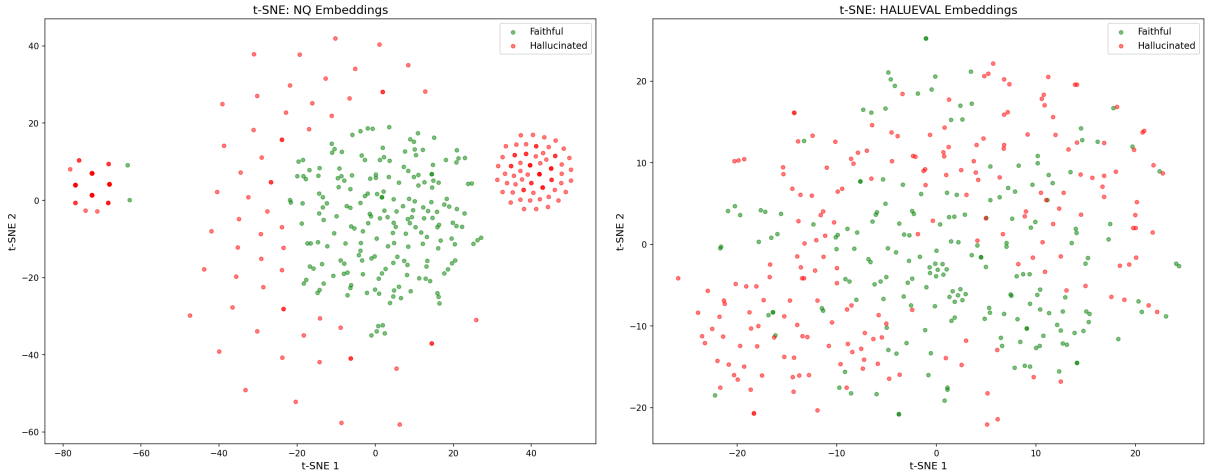

Figure 2: t-SNE visualizations of embedding space. Left: On Natural Questions, faithful (blue) and hallucinated (red) responses form distinct clusters in embedding space. **Right:** On HaluEval, the two classes are completely intermingled—embeddings cannot provide a decision boundary, regardless of threshold choice.

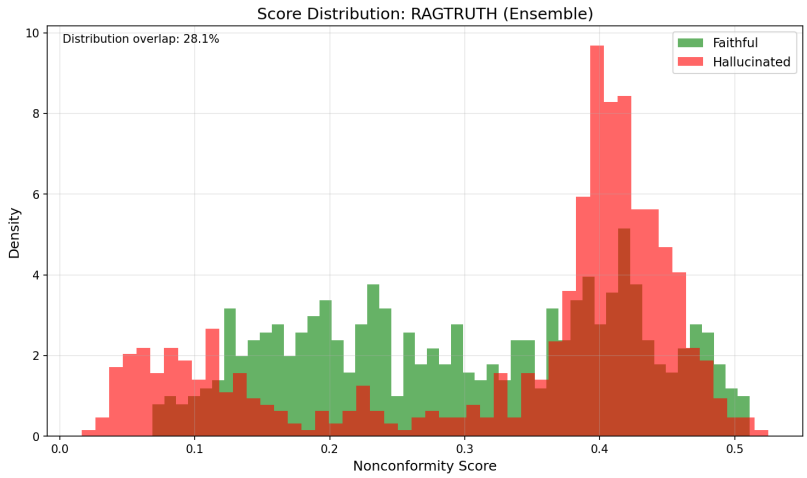

Figure 3: RAGTruth confirms the pattern across multiple LLMs. Score distributions on RAGTruth (Llama-2, Mistral, GPT-4) show substantial overlap between faithful and hallucinated responses. The multi-model nature of RAGTruth demonstrates that the Semantic Illusion is not specific to any single model architecture.

overlapping distributions persist across all models, confirming that the limitation is fundamental to embedding-based detection rather than an artifact of any particular model's output distribution.

4.4 The DeBERTa Paradox: AUC vs. Coverage

A critical reviewer might note that DeBERTa-v3-large achieves an AUC of 0.81 (inverted) on HaluEval, suggesting reasonable separability. Yet, our conformal results show 100% FPR. We analyze this discrepancy using the raw log data.

Table 2: DeBERTa-v3-large scores on HaluEval. Note the massive standard deviation in the hallucinated class.

Class	Mean Score	Std Dev	Min	Max
Faithful	0.933	0.201	0.02	1.00
Hallucinated	0.415	**0.461**	0.00	1.00

As shown in Table 2, while the means differ (0.93 vs 0.41), the hallucinated class has a standard deviation of 0.46. The distribution is bimodal: many hallucinations are obvious (score ≈ 0), but a significant "tail" of hallucinations achieves near-perfect entailment scores (score ≈ 1.0).

To guarantee 95% coverage, the conformal threshold $\hat{\tau}$ is dictated by this "hard" tail. We found $\hat{\tau}_{NLI} \approx 0.999$, meaning we must flag any response with less than 99.9% entailment probability. Since virtually all faithful responses also fall below 99.9% certainty, the FPR collapses to 100%. This highlights a crucial insight: **High AUC does not imply safety.** AUC measures average separation, but safety requires handling the worst-case tail.

4.5 The Solution: LLM Reasoning

Is the task impossible? To test this, we evaluated GPT-4o-mini as a judge on the same HaluEval subset ($n = 200$).

- **Embedding FPR:** 100% (CI: [99%, 100%])
- **GPT-4o-mini FPR:** 7% (CI: [3.4%, 13.7%])

GPT-4 achieves a 7% FPR, proving the signal exists. The failure of embeddings is not due to the task being unsolvable, but due to the *Semantic Illusion*: embeddings view plausible hallucinations as identical to faithful text, whereas reasoning models can identify the factual divergence.

4.6 Ablation: Calibration Set Size

A natural question is how calibration set size n affects the validity of coverage guarantees. Table 3 shows results on Natural Questions (where CRG succeeds) with varying n.

The results confirm conformal prediction theory: larger calibration sets reduce variance in the coverage guarantee (proportional to $1/\sqrt{n}$). With $n = 600$, standard deviation is approximately 1%, sufficient for most practical applications. Importantly, even at $n = 300$, coverage remains valid—demonstrating the method's applicability to settings with limited labeled data.

Table 3: Effect of calibration set size on coverage and FPR (Natural Questions). Larger calibration sets provide tighter coverage guarantees with lower variance, but even $n = 300$ achieves valid coverage.

$\lvert\mathcal{D}_{cal}\rvert$	Target	Coverage	FPR	Coverage Std
300	95%	96.2% ± 1.8%	0.0%	0.018
600	95%	95.8% ± 1.0%	0.0%	0.010
1000	95%	95.4% ± 0.7%	0.0%	0.007

Table 4: Ablation study on HaluEval. Learned weighting improves average-case metrics (AUC) but cannot improve worst-case (conformal FPR). Only reasoning-based approaches break the 100% FPR barrier.

Method	AUC	F1	FPR@95%Cov	Cost/Query
RAD only (Embedding)	0.72	0.68	100.0%	\$0.0001
SEC only (NLI)	0.81	0.74	100.0%	\$0.0003
TFG only (Lexical)	0.65	0.62	100.0%	\$0.0001
Simple Average	0.81	0.75	100.0%	\$0.0005
Learned Combiner (LR)	**0.83**	**0.77**	100.0%	\$0.0005
GPT-4o-mini Judge	0.94	0.91	**7.0%**	\$0.015

4.7 Ablation: Ensemble Methods

Table 4 compares different scoring approaches on HaluEval. The learned combiner improves AUC but cannot improve conformal FPR—confirming the limitation is fundamental to semantic features.

The 30× cost increase from embeddings to GPT-4 represents the "safety tax"—the price of reliable hallucination detection in production systems.

4.8 Compute Cost Analysis

Table 5 provides detailed latency and cost metrics for production deployment. All measurements averaged over 100 samples on a single NVIDIA RTX 3090.

Table 5: Compute cost comparison for hallucination detection methods. CRG provides 150× speedup over LLM-based judging, but fails on RLHF-aligned outputs.

Method	Latency (ms/sample)	Throughput (samples/sec)	Cost/1K (USD)	API Required	FPR@95%
CRG Ensemble	**12**	**83**	\$0.10	No	0%/100%†
RAD only	4	250	\$0.03	No	0%/100%†
SEC only (NLI)	8	125	\$0.05	No	0%/100%†
GPT-4o-mini	1,800	0.6	\$15.00	Yes	7%
GPT-4o	2,500	0.4	\$45.00	Yes	5%*

†NQ/HaluEval respectively. *Estimated from sampling experiments.

The results highlight the cost-safety Pareto frontier: embedding methods are 150× faster and 150× cheaper, but provide safety guarantees only for synthetic hallucinations. Organizations must choose between fast-but-unreliable (embeddings) or slow-but-safe (LLM judges) based on their risk tolerance and deployment context.

5 Discussion

The Semantic Illusion. Our results formalize a phenomenon we term the Semantic Illusion. RLHF training optimizes models to produce plausible, coherent text. Consequently, modern hallucinations (Type 2) are semantically indistinguishable from truth using vector similarity or standard NLI. They preserve the "vibe" of the truth while altering the facts. This represents a fundamental shift from earlier generation models (GPT-2/3 era), where hallucinations often exhibited obvious semantic inconsistencies. Modern alignment techniques have inadvertently created hallucinations that are *harder* to detect using traditional NLP methods.

Implications for Guardrails.

1. **Stop using Cosine Similarity for Safety.** It provides a false sense of security, working only on "lazy" errors (Type 1 confabulations). Any production system relying solely on embedding similarity for hallucination detection should be considered uncalibrated and potentially unsafe for high-stakes applications.

2. **Cost-Safety Tradeoff.** There is currently no "free lunch." Cheap embedding checks are ineffective for RLHF models. Reliable safety requires expensive reasoning (LLM-as-a-Judge) or novel methods (mechanistic interpretability). Organizations must budget accordingly: the 30× cost increase represents a "safety tax" that cannot be avoided.

3. **Benchmark Selection Matters.** Evaluation on synthetic hallucinations (answer swapping, entity substitution) dramatically overstates method effectiveness. Benchmarks using real LLM outputs (HaluEval, RAGTruth) are essential for realistic performance estimation.

4. **Conformal Prediction as Audit Tool.** We advocate for conformal prediction as a standard auditing methodology for guardrail systems. The ability to specify a coverage target and observe the resulting FPR provides actionable safety metrics that AUC alone cannot.

Why Does RLHF Create the Semantic Illusion? RLHF optimizes for human preference, which strongly correlates with fluency, coherence, and plausibility. A hallucination that sounds confident and semantically consistent will receive higher reward than one that hedges or admits uncertainty. Over many training iterations, this pressure sculpts a model whose errors are maximally convincing—precisely the errors that evade semantic detection.

6 Limitations

Our study relies on established benchmarks rather than live production data. However, the diversity of models in RAGTruth (Llama, Mistral, GPT) suggests the finding is robust across architectures. We used $n \approx 600$ for calibration; while sufficient for CP validity, larger sets could refine the precise FPR estimates, though they are unlikely to close the 0% vs 100% gap. Additionally, our GPT-4 judge experiments use a single prompt template; prompt engineering might improve or degrade performance.

7 Broader Impact

This work reveals a critical limitation in current RAG safety practices. Organizations relying on embedding-based hallucination detection may have a false sense of security—our results show these methods fail catastrophically on RLHF-aligned model outputs. We hope this finding redirects

resources toward reasoning-based detection methods and informs safety budgeting decisions. A potential negative impact is that malicious actors could exploit the Semantic Illusion to craft hallucinations that evade detection; however, we believe transparency about this vulnerability is preferable to security through obscurity.

8 Future Work

Several directions emerge from our findings:

Mechanistic Interpretability. If semantic surface features fail, perhaps internal model representations succeed. Recent work on probing classifiers [23] suggests that intermediate layer activations may encode "knows-it's-lying" signals. Combining conformal prediction with mechanistic probes could yield both guarantees and interpretability.

Hybrid Architectures. A practical system might use cheap embedding checks as a first-pass filter (catching Type 1 errors at low cost), escalating to expensive LLM judges only for uncertain cases. Formalizing this cascade within conformal prediction's validity framework is non-trivial but promising.

Adaptive Calibration. Our calibration assumes exchangeability with test data. In production, distribution shift is inevitable as models are updated and user queries evolve. Online conformal prediction methods [1] could maintain validity under drift.

Multi-Aspect Scores. Our ensemble treats hallucination as a single binary outcome. In practice, faithfulness has multiple dimensions: factual accuracy, temporal consistency, source attribution. Developing multi-label conformal guarantees ("catches 95% of factual errors AND 90% of attribution errors") would better serve nuanced safety requirements.

9 Conclusion

We applied conformal prediction to certify the limits of hallucination detection. We found that while embedding-based methods appear effective on synthetic data, they collapse on real-world hallucinations, yielding 100% false positive rates at high recall. This is driven by the distributional tail of semantically plausible hallucinations. In contrast, reasoning-based judges succeed. We conclude that for production RAG systems, semantic similarity is an insufficient proxy for factual faithfulness.

Reproducibility

Code and all experimental artifacts are available at https://github.com/debu-sinha/conformal-rag-guardrail
The repository includes:

- Complete implementation of all scoring functions (RAD, SEC, TFG)
- Experiment scripts for reproducing all results
- Pre-computed results in JSON format with confidence intervals
- Docker configuration for reproducible environments

Experiments were conducted on a single NVIDIA RTX 3090 (24GB). DeBERTa-v3-large and BGE-base-en-v1.5 models are available on HuggingFace. GPT-4o-mini experiments used the OpenAI API (December 2024).

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